

MATH9102: Fundamentals of Data Analysis

Week 4: Comparing Two Groups

Parametric, Non-Parametric and Categorical

Today

1. **Before You Test** — where this sits, and the map of tests
2. **How the t -Test Works** — the statistic, its distribution, its assumptions
3. **The Worked Example** — perceived stress by parenthood, end to end
4. **When Normality Fails** — Mann–Whitney U
5. **Two Categorical Variables** — chi-square of independence
6. **Reporting**

Three tests, one question, one six-step procedure.

Important Note

The work is in **justifying** the test and **interpreting** the result, not in running it.

Before You Test

Where This Week Sits

Before you build a model, you establish that there is something to model.

If you assert a *relationship*

- Look for evidence of association
- Judge its strength and direction
- That was last week — correlation

If you assert a *difference between groups*

- Look for evidence of a difference
- Judge its size
- That is this week

And remember what you are working with

You almost never have the population. You have a **sample**, and every conclusion is a statement about the population made through it. Whether the sample represents the population is a question no test can answer for you.

The Map of Group-Comparison Tests

Two questions decide the test: how are the groups formed, and what kind of variable is the outcome?

	Independent samples two different sets of people	Related samples the same people, twice
Interval / parametric	Independent-samples <i>t</i> -test	Paired-samples <i>t</i> -test
Ordinal / non-parametric	Mann–Whitney <i>U</i>	Wilcoxon signed-rank
Categorical	Chi-square of independence	McNemar's test

This week

The left-hand column, all three rows. The right-hand column is named so you recognise it when your own data are paired — which is a design question, not a statistical one.

Measurement Level Drives the Choice

Interval (and ratio)

- The gap between consecutive points on the scale means the same thing everywhere on it
- Differences and means are meaningful
- Temperature; a summed questionnaire scale treated as continuous

Ordinal

- The categories have an order, but the gaps are not comparable
- “Very good → very bad” coded 1–5
- You can rank; you cannot sensibly average

Watch out

Measurement level is a claim about what your numbers *mean*, and it is yours to make and defend. Software will happily average a Likert item for you. It is not the software's decision.

The Six Steps

You will use these every week from now on.

1. **State** the hypothesis pair
2. **Inspect** the variables
3. **Check** the assumptions, and decide
4. **Run** the test
5. **Report** the result, with an effect size
6. **Conclude** about the hypothesis

Where the effort goes

Step 4 is one line of code. Steps 2, 3, 5 and 6 are the work.

How the t -Test Works

The Logic of a Two-Sample t -Test

The test compares two sample means and asks how surprising their difference is.

If both samples came from one population

- Their means should be similar
- Any gap is sampling variation

If the gap is larger than sampling variation explains

- The samples are likely drawn from two *different* populations
- That is what “a significant difference” claims

The whole idea in one line

Judge the difference between the means **against the difference you would expect by chance**. That ratio is t .

The Test Statistic, and Its Denominator

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\text{SE}(\bar{x}_1 - \bar{x}_2)}$$

The numerator is the effect you observed — the raw gap between the group means, in the units of the outcome.

The denominator is the standard error: the standard deviation of the *difference between means* you would see across repeated samples.

$$\text{SE} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

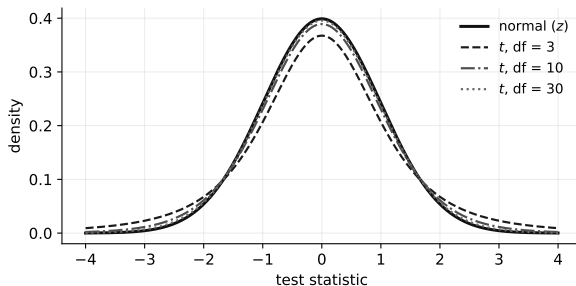
Reading the standard error

Large relative to the means: sample means vary a lot from sample to sample, so this one may not represent the population well.

Small: most samples would give much the same mean, so this one is likely to be an accurate reflection.

Degrees of freedom count the observations free to vary: $n_1 + n_2 - 2$ for the pooled version, and a fractional value for Welch's.

The t Distribution

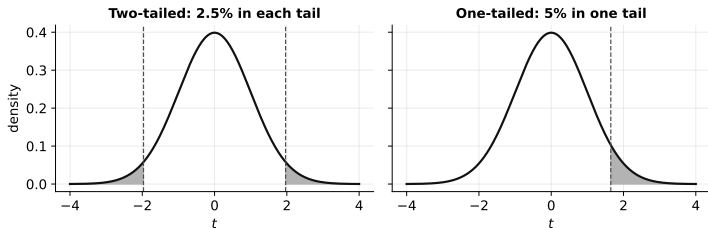


- t has **heavier tails** than the normal — the price of estimating the SD from the sample
- The fewer the degrees of freedom, the heavier
- By a few hundred observations they are indistinguishable

The tables you will not need

Textbooks have you look t up in a table of critical values. Your software computes the exact tail probability instead — the p -value, which says more than “bigger than 1.96”.

One Tail or Two



Two-tailed — the default. $H_0: \mu_1 = \mu_2$; $H_1: \mu_1 \neq \mu_2$. A difference in *either* direction counts, so the 5% is split.

One-tailed. $H_0: \mu_1 \leq \mu_2$; $H_1: \mu_1 > \mu_2$. All 5% on one side — easier to reach, and it commits you to ignoring a difference the other way.

Watch out

Choose the direction *before* you see the data, or use two tails. In this module: two tails unless you justify otherwise.

The Rest of the *t*-Test Family

Independent samples

- Two different groups of people
- Compares their two means
- **This week**

Paired samples

- The same people measured twice
- Compares the mean of the *differences* against zero

One sample

- One group
- Compares its mean against a known or hypothesised value

Watch out

Running an independent-samples test on paired data throws away the pairing, which is usually where the precision was. The test cannot detect this: the numbers arrive in two columns either way. **You have to know your design.**

Assumptions of the t -Test

What it assumes

1. The outcome is measured at least at interval level
2. The outcome is approximately normal **within each group**
3. The groups have roughly equal variances
4. Observations are independent of one another

How much each one matters

- (1) is your judgement, not a test
- (2) matters least when the groups are large and similar in size
- (3) we simply stop assuming — see Welch, shortly
- **(4) is the one with no repair.** It is fixed by the study design, and it is fatal

Robustness is not permission

With large, roughly balanced groups the t -test tolerates a good deal of non-normality. That is a reason not to panic — not a reason to skip looking.

Stand Up — 5 minutes

Five minutes. Nothing to type — nothing has been analysed yet.

Food for thought

- A comparison you would actually want to make — your own data, or your project
- Which row of the map it falls in: how are the groups formed, and what kind of variable is the outcome?
- Which of the four assumptions you could defend, and which you could not

Why now

Everything after this is one worked example. It is easier to follow with a question of your own already in mind.

The Worked Example

Step 1: A Worked Question

Hypotheses

H₀: There is no difference in perceived stress between respondents who have children and those who do not.

H₁: There is a difference in perceived stress between respondents who have children and those who do not.

Outcome

- tpstress — Perceived Stress Scale total
- Continuous, 10 items

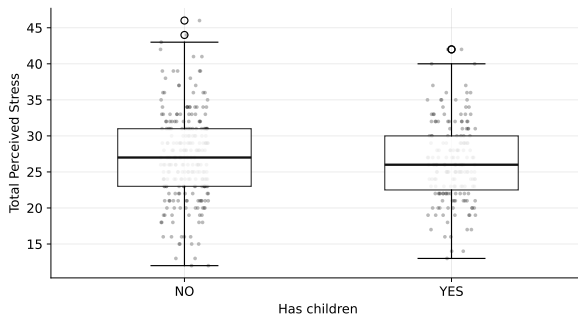
Grouping variable

- child — has children, yes or no
- Binary categorical

Decision rule

- Two-tailed
- $\alpha = .05$, fixed *now*, not after seeing p

Step 2: Look Before You Test



What to read off

- Group sizes
- Spread within each group
- How much the groups overlap

Before any test

If the boxes overlap almost entirely, expect a small effect whatever the p -value turns out to be.

Step 2: Describe Each Group

```
# Step 2. Describe the grouping variable, then the outcome within each group.  
# Group sizes come first: an effect size means little without them.  
m9.frequency(survey, "child")  
m9.describe_by(survey, "tpstress", "child")  
  
# What a complete-case analysis costs, on the model variables only -  
# never on the whole dataframe.  
missing = m9.missingness(survey, ["tpstress", "child"])  
print(missing.attrs["summary"])
```

You need, per group

- n , and how many are missing
- Mean and standard deviation
- Enough shape to justify step 3

Watch out

Drop missing rows *after* choosing the model variables. Complete-case deletion across all 134 columns here costs a quarter of the sample, most of it to variables you never model.

Step 3: Normality — Of What, Exactly?

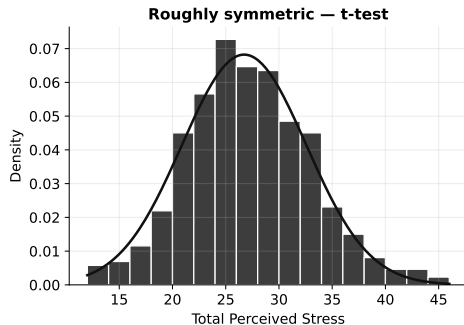
The t -test assumes the outcome is roughly normal *within each group*.

Watch out

Checking the *pooled* distribution mixes the groups together. If the group means differ, the pooled distribution can look bimodal or skewed even when each group is perfectly normal.

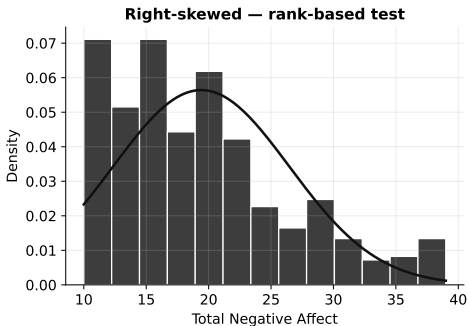
Assess it **visually first**. A histogram and a Q–Q plot per group tell you more than any test statistic.

Two Outcomes, Two Decisions



Left: Perceived Stress

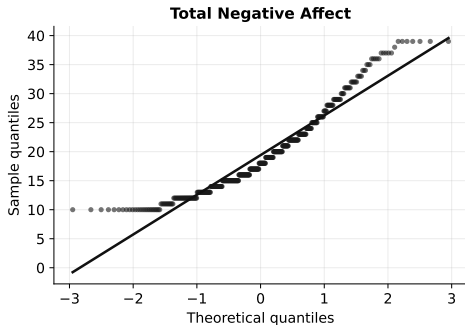
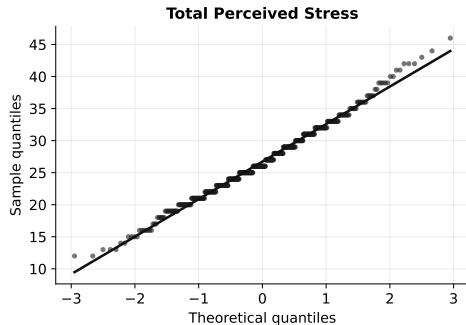
- Roughly symmetric
- A *t*-test is appropriate



Right: Negative Affect

- Floor at the bottom of the scale, tail to the right
- Use a rank-based test

The Same Contrast on Q-Q Plots



How to read a Q-Q plot

Points on the line means normal. Curvature at one end means skew. This is the plot to read when you are deciding which test to use.

Why Not Just Test for Normality?

Shapiro–Wilk, and standardised skew and kurtosis, all scale with sample size.

- At $n = 30$ they miss departures that matter
- At $n = 400$ they flag departures that do not

So report them as **descriptions of shape**, and let the plots decide.

Watch out

If you find yourself writing “the test was significant, but we proceeded anyway”, the test was never making the decision. Say what the plots showed instead.

Equal Variances: Don't Gate on a Test

A common procedure runs Levene's test, then picks a pooled or separate-variance t -test depending on the result.

Our convention

Always use Welch's t -test.

Why

- Choosing your test from another test's outcome distorts the error rate of the one you care about
- $p = .049$ and $p = .051$ would give different analyses of one dataset
- Welch costs almost nothing when variances really are equal

What to do instead

- Still *look* at Levene's test
- Still report the group SDs
- Describe what you saw — don't branch on it

Step 4: Welch and Pooled, Side by Side

```
# Welch is the default. The pooled test is shown only for comparison -  
# never choose between them on the outcome of a variance test.  
welch = stats.ttest_ind(no, yes, equal_var=False)  
pooled = stats.ttest_ind(no, yes, equal_var=True)  
  
print(f"Welch: t = {welch.statistic:.4f}, p = {welch.pvalue:.4f}")  
print(f"Pooled: t = {pooled.statistic:.4f}, p = {pooled.pvalue:.4f}")
```

On this example

The two agree. That is the usual case, and it is why defaulting to Welch is safe rather than cautious.

Step 5: What the Output Is Telling You

Read, in this order

1. The **mean difference** — in the units of the outcome. Is it a difference anyone would notice?
2. The **confidence interval** on that difference. If it comfortably spans zero, say so
3. The **test statistic**, its degrees of freedom, and p
4. The **effect size**

The order matters

Students who read p first tend to stop there. The mean difference is the finding; p only says whether the data can support claiming it.

A perceived-stress scale runs over tens of points. Ask what a difference of less than one point on it would mean to anybody.

Step 5: Always Report an Effect Size

A p -value answers: *could this difference plausibly be noise?*

It does **not** answer: *is this difference big enough to care about?*

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}} \quad s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Cohen's d is the difference expressed in standard deviations. Rough benchmarks: 0.2 small, 0.5 medium, 0.8 large. Benchmarks, not verdicts.

Watch out

You may meet the shortcut $d = 2t/\sqrt{\text{df}}$. It assumes the two groups are the *same size*. Use the pooled formula.

Two Effect Sizes, and One Shortcut

```
# The pooled form works for unequal group sizes. Use it.  
d = m9.cohens_d(no, yes)  
print(f"Cohen's d = {d:.3f} ({m9.interpret(d, 'd')}")  
  
# The textbook shortcut assumes n1 == n2, so it is an approximation here.  
approx = 2 * pooled.statistic / np.sqrt(len(no) + len(yes) - 2)  
print(f"2t/sqrt(df) = {approx:.3f}")  
  
# Eta squared: the share of variance in the outcome the grouping accounts for.  
print(f"eta squared = {m9.eta_squared_from_t(pooled.statistic, pooled.df):.4f}")
```

Cohen's d — the gap in standard deviations.

0.2/0.5/0.8.

Eta squared — the share of variance in the outcome that group membership accounts for.

.01/.06/.14.

Which one

They answer one question in different units, and disciplines differ on which they expect. Report one, name it, stay consistent.

A Number Nobody Checked

A published Cohen's d of 0.06 — half of what the formula printed beside it gives.

What happened

- The page printed the formula $2t/\sqrt{df}$
- The arithmetic beside it left out the 2
- Every conclusion drawn from it described an effect half the size

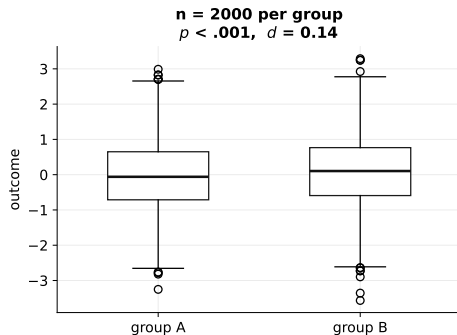
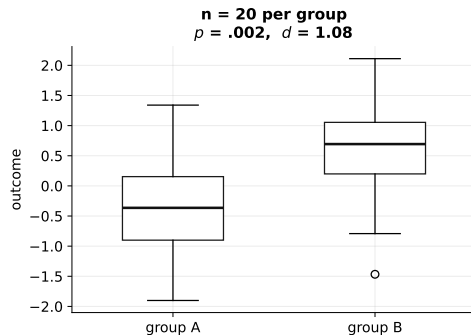
Why it survived

- It was **typed**, once, by hand
- Nothing recomputed it when the analysis was re-run
- In prose, a wrong number looks exactly as authoritative as a right one

The lesson this module draws

Every statistic you report is generated from the data by code. Not because typing is lazy — because a typed number goes stale silently, and the silence is the problem.

Significance Is Not Size



Both significant, and only one worth reporting

On the left the groups barely overlap. On the right they are nearly identical, and the difference is detectable only because there are 2000 people per group — the smaller p belongs to the negligible effect.

Break — 30 minutes: Run the t -Test

20 minutes on the analysis, then 10 for tea.

In `notebook.ipynb`

- **Part 1**, all six steps, top to bottom
- Read each output before running the next cell

Then in `exercise.ipynb`

- Question 1 — the same six steps on a question you have not seen

Then stop

The last ten minutes are tea, not catch-up. If Part 1 did not finish, bring it to the end of the session.

Think about

The mean difference is under one point on a scale running over tens of them. What would you tell someone who asked whether it mattered?

Where You Should Be: the t -Test

You should now have

- A generated report sentence — every number in it produced by the code, none typed
- An effect size you can describe in words, not just quote
- A conclusion that says what the evidence does *not* establish

The two that catch people

Normality checked **within each group**, not on the pooled outcome. And Levene's test *described*, never used as a gate.

When Normality Fails

When Ranks Are the Right Tool

Reach for a rank-based test when

- The outcome is clearly skewed or bounded
- The outcome is ordinal
- The sample is small enough that you cannot see the shape

It still assumes

- A random sample
- **Independent** observations
- Each case counted once — unless the design is genuinely repeated measures, which needs a different test

“Non-parametric” does not mean “assumption-free”

It means the test does not assume a particular *distribution*. The design assumptions are exactly as strict as before.

A Worked Question: Two Drinks

Twenty students at a social event; ten drank Drink X, ten drank Drink Y. Depression was measured on the Beck Depression Inventory the next morning (Sunday) and again midweek (Wednesday).

Hypotheses

H₀: Depression scores do not differ between the two drinks.

H₁: Depression scores differ between the two drinks.

Ten observations per group is far too few to judge normality from a histogram, and a normality test at $n = 10$ has almost no power.

So the decision is made on design

A bounded questionnaire score, ten per group. Compare ranks.

How Ranking Works

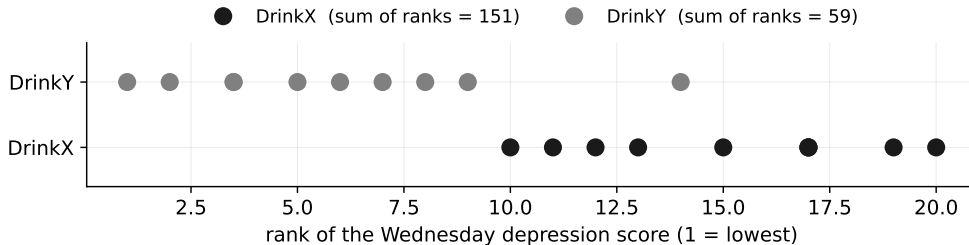
Pool both groups, sort, and number from the bottom.

- Lowest score gets rank 1, next gets rank 2, and so on
- **Tied scores share the average** of the ranks they would have occupied
- Add up the ranks within each group
- If the groups do not differ, high and low ranks are scattered evenly between them
- If one group holds most of the high ranks, that is the signal
- The analysis is done on the **ranks**, not on the original scores

Watch out

Ties are ordinary in questionnaire data. They cause a warning about exact p -values, not an error — and they are *not* missing data.

What the Ranks Look Like



Every Wednesday score, replaced by its rank among all twenty. One group holds nearly all the high ranks.

This picture is the test

Everything Mann–Whitney does is a formal statement about how separated these two rows are.

The rank-based counterpart of the independent-samples t -test.

What it does

- Pools the observations and ranks them
- Asks whether one group's ranks sit systematically higher
- Describe the groups with **medians and IQRs**, not means and SDs

Its relatives

- Wilcoxon *rank-sum* is the same test under another name; your software may report either
- Wilcoxon *signed-rank* is the paired version — a different test

It also works on normal data

Just less efficiently than a t -test. Choosing it when a t -test is defensible costs you power, so justify the choice either way.

The U Statistic

U counts how often a member of one group outranks a member of the other.

- Take a group's rank sum, subtract the smallest sum that group could possibly have had
- What remains is U for that group
- Software may report the smaller U , the larger, or W — read your own documentation
- Small samples: U is compared against exact critical values
- Larger samples — roughly $n > 30$ per group — U is close enough to normal to use a z approximation
- Ties are handled by correcting that approximation

Watch out

Because different tools report different members of the same family, always report **which statistic** you quote alongside its value.

Running It, and Sizing It

```
# Mann-Whitney U compares ranks, so describe the groups with medians and IQRs.
x = bdi.loc[bdi.drink == "DrinkX", "bdiwed"]
y = bdi.loc[bdi.drink == "DrinkY", "bdiwed"]

mw = stats.mannwhitneyu(x, y, alternative="two-sided")

# Two different quantities are both written r for this test. Say which you used.
print(f"rank-biserial r = {m9.rank_biserial(x, y):.3f}")
print(f"Rosenthal's r = {m9.wilcoxon_r(x, y):.3f}")
```

Rank-biserial r

- Computed from U directly
- $(r + 1)/2$ is the probability that a random member of one group outscores one from the other

Rosenthal's $r = |z|/\sqrt{N}$

- The older convention, and still common in the literature
- Numerically different — **not** a disagreement, a different quantity

What the Test Actually Claims

Mann–Whitney compares whole distributions, not medians.

A significant result supports

“Scores in group A tend to be higher than in group B.”

It does not, on its own, support

“The median of group A is higher than the median of group B.”

(That reading needs the two distributions to have a similar shape.)

This is why you report medians and IQRs *as description* and phrase the conclusion in terms of tendency. The distinction costs one word and is the difference between a defensible claim and an overstated one.

Stand Up — 5 minutes: Halfway

Five minutes. Two tests done, one to go.

Food for thought

Both tests so far compared a *measurement* between two groups — one on means, one on ranks.

The last one has no measurement in it at all. Only counts. So what would “no difference” even mean?

Still the same six steps

Only step 4 changes. The hypothesis, the inspection, the assumption, the effect size and the conclusion are the work every time.

Two Categorical Variables

Why Frequencies, Not Means

A mean or median of a categorical variable is meaningless.

- The numbers attached to categories are **labels**, chosen arbitrarily
- Their average depends on how many people happen to be in each category, and on which label you assigned to which category
- Recode the labels and the “mean” changes

So we analyse counts

Two nominal variables are **independent** when knowing which category a case falls into on one tells you nothing about its probability of falling into any category of the other.

A Worked Question: Bullying

A national school survey, asking two yes/no questions.

- ubullsch — have you ever been bullied in school?
- ubulloth — have you ever taken part in bullying other students?

The shape of the data

Two binary variables give a 2×2 table, which has one degree of freedom: fix one cell and the margins, and the other three follow.

Hypotheses

H₀: Being bullied at school and bullying others are independent.

H₁: Being bullied at school and bullying others are associated.

Chi-Square of Independence

The test compares the counts you **observed** against the counts you would **expect** if the variables were independent.

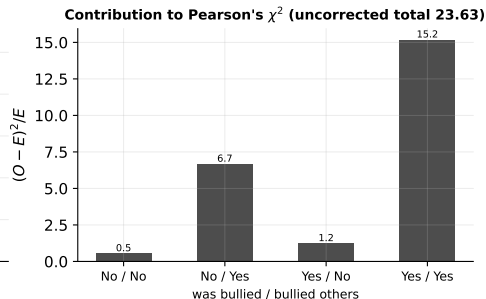
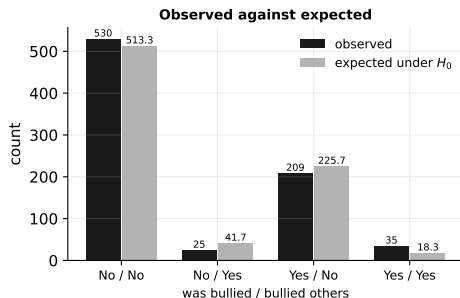
$$\chi^2 = \sum \frac{(O - E)^2}{E} \qquad E_{ij} = \frac{(\text{row } i \text{ total}) \times (\text{col } j \text{ total})}{N}$$

- Each cell contributes $(O - E)^2/E$
- A large total means the data do not look like independence
- Compare against the χ^2 distribution on the table's degrees of freedom

The expected table

E is not a guess. It is the table you would get if the row proportions were identical across every row — built from the margins alone.

Observed Against Expected

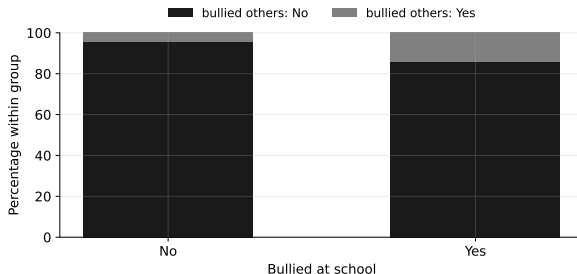


More students than expected were *both* bullied and bullied others; fewer than expected bullied others without having been bullied.

Read the right-hand panel

The two *small* cells contribute nearly all of χ^2 ; the largest cell contributes almost none.

Reading a Contingency Table



Why row percentages

Percentages *within rows* answer the useful question: of the people in this group, what proportion did the other thing?

Watch out

Say which way you percentaged. Row, column and total percentages of one table tell three different stories.

Running It in Python

```
# Chi-square of independence. correction=True applies Yates, which is what you
# want for a 2x2 table; scipy applies it only to 2x2 tables in any case.
table = m9.crosstab(bully.dropna(subset=["ubullsch", "ubulloth"]),
                    "ubullsch", "ubulloth")

chi2, p, dof, expected = stats.chi2_contingency(table, correction=True)

# Check the assumption before you read the result, not after.
print(f"smallest expected count: {expected.min():.1f}")
print(f"Cramer's V = {m9.cramers_v(table):.3f}")
```

Note the order

The expected counts come out of the same call as the test statistic, so there is no excuse for reading the result before checking the assumption it depends on.

Assumptions: It Is the *Expected* Counts That Matter

The rule

- Every **expected** count should be above 5
- Not the observed counts — an observed zero is fine if its expected count is healthy
- Check it; do not assume it

When it fails

- Use **Fisher's exact test** instead — it computes the probability directly rather than approximating it
- Or collapse sparse categories, if that is defensible on substantive grounds rather than convenience

Why the rule exists

χ^2 approximates a discrete count distribution by a continuous one. The approximation is good when the counts are large and poor when they are not — there is nothing magic about 5.

Yates' Continuity Correction

The same approximation problem, addressed a second way.

- Subtract 0.5 from each $|O - E|$ before squaring
- This shrinks χ^2 , and so raises p
- Guards against a Type I error — claiming an association that is not there
- Applies to 2×2 tables only

You can see it in the figure

The bars on the previous figure sum to the *uncorrected* statistic. The corrected one, which is what gets reported, is visibly lower.

Watch out

Yates is contested: many statisticians consider it over-conservative even for small tables. Follow your field's convention, and **say which you applied** — the two answers differ.

Effect Size: ϕ and Cramér's V

ϕ (phi)

- For 2×2 tables
- $\sqrt{\chi^2/N}$
- Benchmarks .1 small, .3 medium, .5 large

Cramér's V

- For any size of table
- Identical to ϕ when the table is 2×2

$k - 1$	small	medium	large
1	.10	.30	.50
2	.07	.21	.35
3	.06	.17	.29
4	.05	.15	.25
5	.04	.13	.22

k is the *smaller* side of the table: $k - 1$ is $\min(r - 1, c - 1)$, not the table's df. A 2×5 table takes the first row, not the fourth.

Watch out

With several hundred observations, even a very weak association reaches significance. The effect size is what tells you whether it matters.

When the Categorical Data Are Paired

Chi-square assumes each person appears once. Sometimes they do not.

McNemar's test

- One binary variable, measured twice — before and after an intervention
- Tests whether the *changes* in one direction outnumber those in the other
- It reads the off-diagonal cells only

Cochran's Q

- The same idea for three or more measurement occasions
- Or three or more matched conditions

The pattern by now should be familiar

Independent groups and repeated measures need different tests at every level of measurement. Which one you need is settled by the study design, before any data are collected.

Break — 30 minutes: Ranks and Counts

25 minutes on the analyses, then 5 to stretch.

In `notebook.ipynb`

- **Part 2** first — Mann–Whitney on the drink study
- **Part 3** — chi-square on the bullying data

Then in `exercise.ipynb`

- Questions 2 and 3, as far as you get

Part 2 before Part 3

Mann–Whitney was taught twenty minutes ago; the chi-square is still on the screen behind you.

Think about

Why the chi-square assumption is about *expected* counts, when the data in front of you are observed ones.

Where You Should Be: Ranks and Counts

You should now have

- A Mann–Whitney result, reported with **medians and IQRs** — not means
- A chi-square result with its **expected counts checked** before the statistic was read
- An effect size beside each, named and interpreted

If something did not run

Keep going — the reporting section needs none of it, and the worked solution is released after this class. Note where it stopped and ask.

Reporting

How to Report a Result

Every result needs all of this:

- What test, and on what variables
- Descriptive statistics for each group, with n — means and SDs for a t -test, medians and IQRs for Mann–Whitney, counts and row percentages for chi-square
- The test statistic, its degrees of freedom, and p
- An effect size, **interpreted**
- A conclusion referring back to the hypothesis

Convention

Report p to three decimals, without a leading zero. Below .001, write $p < .001$.

Report the effect size **even when the result is not significant**.

Generate Your Numbers — Never Type Them

In this module, every statistic in a written report is produced by code.

```
# The report sentence is generated from the data and the result object,  
# so its numbers cannot drift from the analysis.  
result = stats.ttest_ind(no, yes, equal_var=False)  
  
m9.report_ttest(survey, "tpstress", "child", result,  
                outcome_label="Total Perceived Stress",  
                group_labels={"NO": "respondents without children",  
                             "YES": "respondents with children"})
```

Why

- Typed numbers go stale when the data or analysis changes
- Nobody proofreads a number they wrote three weeks ago

How to check your own

Change the data, re-run, and watch every number move. Any that does not was typed.

Two Conclusions You Must Not Draw

A non-significant result

Does *not* mean the groups are the same, and does *not* prove H_0 . It means this sample did not provide sufficient evidence of a difference.

A significant result

Does *not* prove H_1 , and does *not* mean the difference is important. Check the effect size before you write the word “substantial”.

What you should take from this week

You SHOULD be able to

- Choose between the three tests from the variable types, the design and the evidence
- Check normality *within groups*, visually first
- Run and interpret a Welch t -test, a Mann–Whitney U , and a chi-square
- Check the assumption each test actually makes, and say what you saw
- Report an effect size alongside every test, and describe its magnitude
- Write the result so that every number in it was generated, not typed

This week: the notebook works all three tests end to end; the exercise asks you to do the same on variables you have not seen.

That's all for this week